# From Case Retrieval to Normative Material Screening: A Proof-Carrying Audit Protocol for Legal AI Recommendation Systems


**Abstract**

Legal artificial intelligence (AI) case recommendation systems decide which legal authorities become visible before legal reasons are written. They are procedural machines, not search accessories. This article introduces *normative material screening output* as the middle category between reference information and decision reasons: a ranked, filtered or summarized authority set that structures later argument without itself becoming law. The framework assigns procedural status through output effect, system-role caps, failure caps and a six-dimension audit vector covering source support, query trace, authority hierarchy, ranking/counter-material salience, contestability and adoption logging. It is implemented as a provider-agnostic harness that emits proof-carrying status certificates. The validation is adversarial: public retrieval outputs, single-model and cross-engine transcripts, source-supported repairs, source-chain attacks, claim-anchor falsification, workflow-portability checks, issue-family and multi-axis holdouts, formal invariants, finite-state status-lattice exhaustion, policy replay, policy-family robustness, construct-operationalization coverage, threat-model coverage, certificate tamper-resistance and coding-uncertainty analysis all test whether status can be reproduced by a weaker rule. The result is categorical: procedural status is not reducible to recall, retrieval metrics, authority coverage, fluent generation, total score, source binding, human-review posture, workflow architecture or model identity. In the finite high-status lattice, the best partial substitute admitted 672 false positives; the full screening predicate admitted 0 false positives and 0 false negatives. The contribution is a proof-carrying audit protocol for deciding when case-recommendation outputs remain reference information, become professional support, qualify as normative material screening, enter decision reasons or lose external legal effect.

**Keywords:** legal information retrieval; case recommendation; legal AI auditing; contestability; output evidence packets; normative material screening

# 1 Introduction

Legal artificial intelligence (AI) systems increasingly decide what law becomes visible. Case recommenders, generated research assistants and court-facing authority reports set the field of authorities from which reasons are built. The central risk is not hallucinated citations alone; it is authority selection without procedural status.

This article's claim is direct: procedural status should follow auditability, not intelligence. A fluent answer with weak sources remains reference information. A high-scoring ranked list with no counter-material pathway remains professional support. Only a reconstructable and contestable authority-screening chain can become procedural material. The missing category is *normative material screening*: outputs that filter, rank or summarize legal authorities without becoming legal sources or decision reasons.

The paper therefore asks three operational questions:

RQ1: How should AI-generated legal outputs be classified according to their procedural effects?

RQ2: What verifiability and contestability requirements are needed for each output class?

RQ3: How can these requirements be operationalized and stress-tested across implementation, public-output, model-output, source-grounding, adversarial, recoding and uncertainty settings?

The resulting framework has two dimensions. The output-effect dimension classifies what the output does in the legal process. The system-role dimension classifies how the system enters that process. The two dimensions are then linked to verifiability, contestability and accountability requirements. The model is intentionally output-first: it asks when a legal AI output may move from internal reference to procedural material, and when it must be downgraded or withdrawn.

The contributions are three. First, the article names and formalizes normative material screening as a distinct procedural object in AI and Law. Second, it gives that object an explicit status function: output effect, role cap, audit vector, mandatory gates and failure caps jointly decide the maximum procedural status. Third, it supplies an executable validation matrix: 264 scenario packets, 697 embedded records or outputs, and 8,023,761 validation operations across public retrieval, single-model and cross-engine transcripts, source-grounding repairs, adversarial source-chain attacks, claim-anchor falsification, workflow-portability checks, issue-family and multi-axis holdouts, policy-family robustness, gate ablations, policy-mutation tests, review-provenance tests, formal invariants, finite-state status-lattice exhaustion, construct-operationalization coverage, threat-model coverage, proof certificates, policy replay, model-identity substitution and coding-uncertainty analysis. The matrix is built to defeat six substitute theories: performance sufficiency, source-label sufficiency, authority-material sufficiency, review-label sufficiency, score sufficiency and model-identity sufficiency. Each substitute fails once the legal-material chain is tested.

To make the artifact interpretable as well as large, the evaluation includes a worked evidence ladder: the same Loper Bright output moves from reference information, to normative material screening, to decision-support reason, or to no external legal effect solely as source bindings, contestability, adoption and authorization records change.

AI-based case recommendation is the core case because it sits at the boundary between legal information retrieval and legal reasoning. Case recommendation systems are central to case-based reasoning and legal information retrieval, both long-standing areas of AI and Law research (Ashley, 1992; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). They are also procedurally sensitive. In common-law precedent search, civil-law court-decision databases, administrative adjudication systems, arbitral award repositories and online dispute resolution (ODR) platforms, ranking and omission can affect the legal materials available for argument. The framework is therefore jurisdiction-parameterized rather than jurisdiction-neutral: each legal system supplies its own authority hierarchy, while the verifiability problem remains shared.

# 2 Related Work and Gap

## 2.1 Legal information retrieval and case-based reasoning

Legal information retrieval and case-based reasoning both show why case recommendation cannot be evaluated as generic text ranking. Legal materials are constitutive as well as descriptive: statutes, precedents, administrative rules and court decisions may structure the legal domain itself. Legal retrieval must therefore account for hierarchy, temporality, authority, citations and professional use rather than reduce relevance to algorithmic similarity (Turtle, 1995; Rissland and Daniels, 1995; van Opijnen and Santos, 2017). Case-based reasoning reaches the same point from legal reasoning: cases matter through legally relevant similarities, differences, factors, values and precedential relations (Ashley, 1992; Bench-Capon and Sartor, 2003). The procedural risk is that a system may make some authorities salient and others invisible while presenting that ordering as ordinary search.

Legal argumentation research supplies the adjacent reasoning vocabulary. Abstract argument acceptability, burden-of-proof models and argumentation accounts of law explain how reasons conflict and how justified conclusions may be drawn (Dung, 1995; Gordon et al., 2007; Prakken and Sartor, 2015). The present article sits one step earlier. It does not ask which argument should win; it asks whether the output that supplies cases, statutes or counter-materials may procedurally enter an argument at all. That placement separates procedural status from downstream legal merits.

## 2.2 Benchmarks and legal retrieval models

Benchmarking work has substantially improved the technical evaluation of legal retrieval. COLIEE defines shared tasks for case-law retrieval, case-law entailment, statute retrieval and statutory entailment, with explicit datasets, runs and leaderboard metrics (Goebel et al., 2024). Neural retrieval systems such as BERT-PLI model paragraph-level interactions between a query case and candidate cases, addressing the length and structure of legal cases more directly than generic ad hoc retrieval models (Shao et al., 2020). Legal language and holding benchmarks such as LexGLUE and CaseHOLD extend evaluation to standardized legal-language understanding and legal holding selection (Chalkidis et al., 2022; Zheng et al., 2021). Recent retrieval-augmented legal benchmarks, including reasoning-focused legal retrieval and CLERC, move closer to professional research tasks by connecting retrieval to downstream legal analysis and citation support (Zheng et al., 2025; Hou et al., 2025). LegalBench further extends the evaluation agenda to a broader set of legal reasoning tasks for large language models (Guha et al., 2023).

These benchmarks answer the performance question. This article answers the status question: after an output exists, what procedural position may it claim? COLIEE, LegalBench, CLERC and retrieval benchmarks ask whether a system retrieves, entails, cites or reasons correctly under task conditions. Procedural qualification asks whether the ranked set may be attached to a court report, relied upon in a professional opinion or incorporated into a decision reason. That question turns on corpus records, authority labels, counter-authority handling, contestability and adoption logs.

## 2.3 Generated legal outputs and source verification

Generated legal interfaces sharpen the procedural problem. A ranked list exposes a result set for inspection; a generated answer may compress selected materials into a fluent narrative and conceal omissions, conflicts or weak source support. Empirical work on legal hallucinations and commercial legal research tools shows that legal language fluency and retrieval augmentation cannot substitute for source verification

(Dahl et al., 2024; Magesh et al., 2025). The audit task is to separate source truth, retrieval relevance, authority status and procedural use. Technical usefulness and procedural eligibility are different properties.

### 2.4 Auditability, explainability and contestability

Algorithmic accountability, AI auditing and contestability scholarship supplies the institutional premise: consequential automated outputs must be reviewable, challengeable and traceable rather than accepted as self-justifying (Kroll et al., 2017; Raji et al., 2020; Mökander, 2023; Wachter et al., 2017; Binns et al., 2018; Citron and Pasquale, 2014; Veale et al., 2018). Case recommendation needs a domain-specific version of that premise because it structures the visibility and ranking of legal authorities without itself deciding a case. Normative material screening is that version.

## 3 Case Retrieval as Normative Material Screening

This article calls such outputs *normative material screening outputs*. The category identifies an intermediate procedural object: an output that filters, ranks or summarizes legal materials that may later support legal argument or decision-making, without itself creating legal authority or deciding rights and duties. Examples include AI-generated lists of similar cases, recommended statutes, ranked precedent sets, extracted holdings and counter-authority reports.

The category rejects two false equations: retrieval success equals procedural permission, and legal usefulness equals a decision reason. Many legal AI outputs shape the preparation of legal reasoning without becoming reasons for a decision. The middle layer is where case recommendation systems actually operate.

## 4 A Two-Dimensional Output Qualification Model

The proposed framework starts from a simple premise: the normative qualification of a legal AI output depends on both what the output does and how the system enters the procedure. These are separate dimensions.

### 4.1 Output-effect dimension

The output-effect dimension classifies the legal effect claimed for the output:

The classification uses four criteria: whether the output is exposed outside the user's internal workspace, whether a professional relies on it in advice or preparation, whether it filters the set of legal authorities available for argument, and whether an authorized decision-maker adopts it as part of a reasoned outcome. These criteria keep procedural status separate from model quality.

1. **Reference information.** The output supports internal search, summarization or drafting. It has no external procedural effect and is not presented as a basis for another person's legal position.

2. **Professional support output.** The output assists a lawyer, expert, mediator or other professional in forming advice. The professional remains responsible for verification and communication.

3. **Normative material screening output.** The output filters, ranks, summarizes or highlights legal authorities or other normative materials. It may shape the materials available for argument, response or judicial consideration.

4. **Decision-support reason.** The output is adopted, summarized or incorporated by an authorized human or institution as part of the reasons for a legal decision, settlement recommendation, administrative action or arbitral outcome.

5. **Failure consequence: no external legal effect.** This cap applies to withdrawal-level failures and unaccountable rights-affecting external use.

### 4.2 System-role dimension

The system-role dimension determines how far an output may travel procedurally. In the executable protocol, $R(o)$ is a role cap: back-office use, disclosed assistance, auditable procedural use and authorized decision support permit different maximum statuses. The same ranked case list may therefore be private reference, professional support, court-facing screening, or a decision reason depending on the role and adoption path. Table 1 summarizes the role question and audit artefact attached to each output class.

## 5 A Verifiability-Based Audit Model

The framework can be expressed as an audit model. Let $o$ be a legal AI recommendation output generated in a particular institutional setting. Its procedural status $PS(o)$ is a function of output effect, system role and an audit vector:

$$PS(o) = g(E(o), R(o), \mathcal{A}(o)).$$

Here $E(o)$ denotes output effect, $R(o)$ denotes system role, and $\mathcal{A}(o)$ denotes a six-part audit vector:

Table 1: Output status and procedural role

| **Output class** | Typical use | Minimum procedural question | Audit artefact required |
| --- | --- | --- | --- |
| **Reference information** | Internal search, summary, first-pass drafting | Can the user verify the source before relying on it externally? | Source record and user-visible citation or document link. |
| **Professional support output** | Lawyer advice, expert preparation, mediator support | Can the professional explain the basis and limitations of the output to the affected person? | Source, query, output and limitation record. |
| **Normative material screening output** | Case recommendation, authority ranking, statute or precedent set selection | Can parties or reviewers reconstruct the corpus, version, ranking logic and omitted counter-materials? | Corpus manifest, ranking record, counter-material record and review log. |
| **Decision-support reason** | Output adopted in a judgment, administrative decision, arbitral award or ODR recommendation | Can the adoption path, human reasoning and contestation record be audited? | Adoption log, human reasons, contestation record and final responsibility node. |
| **No external legal effect** | Unaccountable or withdrawal-level output affecting rights, duties or procedural opportunities | Does the output claim rights-affecting external effect without authorization, reconstructability, contestability or accountability? | Failure record stating the external-effect bar. |

$$\mathcal{A}(o) = (S, Q, H, K, T, L).$$

The six components are source and corpus verifiability ($S$), query and version traceability ($Q$), authority-hierarchy representation ($H$), ranking and counter-material verifiability ($K$), contestability support ($T$), and adoption logging with audit responsibility ($L$). Each component is scored on a three-point ordinal scale: 0 means absent, 1 means partially available, and 2 means procedurally reviewable. Table 2 defines the scoring dimensions. The model is a status-allocation protocol: it determines the highest procedural status that the output may claim given the audit evidence available.

## 5.1 Formal definitions and invariants

Let the procedural statuses be ordered as

$$\mathcal{P} = \{p_0, p_1, p_2, p_3, p_4\},$$

where $p_0$ is no external legal effect, $p_1$ is reference information, $p_2$ is professional support output, $p_3$ is normative material screening output and $p_4$ is decision-support reason. The order $p_0 \prec p_1 \prec p_2 \prec p_3 \prec p_4$ expresses the maximum procedural status an output may claim. Let $\mathcal{A}(o) \in \{0, 1, 2\}^6$ be the audit vector and let $\Pi = (\tau_N, \tau_D, \Gamma, \Delta)$ be a status policy, where $\tau_N$ is the normative-screening threshold, $\tau_D$ is the decision-support threshold, $\Gamma$ is the set of necessary gate predicates and $\Delta$ is the failure-flag cap.

System role is coded as $R(o) \in \mathcal{R}$, where $\mathcal{R}$ contains back-office tools, disclosed assistance tools, auditable procedural tools, authorized decision-support tools and unaccountable external disposition tools. Define a role-cap function

$$c(R) = \begin{cases} p_1, & R = \text{back-office tool}, \\ p_2, & R = \text{disclosed assistance tool}, \\ p_3, & R = \text{auditable procedural tool}, \\ p_4, & R = \text{authorized decision-support tool}, \\ p_0, & R = \text{unaccountable external disposition tool}. \end{cases}$$

Output effect sets the claimed status; system role sets the maximum status available in the procedure. The operational rule then computes the highest procedurally available status from the audit vector, the role cap and the relevant artefact gates. Let $B_A(o)$ denote authority-set completeness, $B_E(o)$ evidence-packet completeness, $B_M(o)$ material claim-anchor completeness, $B_P(o)$ procedural source-tag completeness, $B_C(o)$ counter-material completeness, $B_J(o)$ jurisdiction-profile support, $B_T(o)$ completed review plus an explicit contestability-channel record, $B_D(o)$ completed adoption support and $B_F(o)$ the absence of an active downgrade, suspension or withdrawal flag. For a candidate status $p$, define a gate predicate $m_p(o, \Pi) \in \{0, 1\}$:

$$\begin{aligned}
&m_{p_1}(o) = \mathbf{1}[S \geq 1 \wedge Q \geq 1], \\
&m_{p_2}(o) = \mathbf{1}[m_{p_1}(o) \wedge L \geq 1], \\
&m_{p_3}(o) = \mathbf{1}[S, Q, H, K, T, L \geq 1 \wedge \sum \mathcal{A}(o) \geq \tau_N \\
&\qquad \wedge B_A(o) \wedge B_E(o) \wedge B_M(o) \wedge B_P(o) \wedge B_C(o) \wedge B_J(o) \wedge B_T(o) \wedge B_F(o)], \\
&m_{p_4}(o) = \mathbf{1}[m_{p_3}(o) \wedge \sum \mathcal{A}(o) \geq \tau_D \wedge T = L = 2 \wedge B_D(o)].
\end{aligned}$$

The preliminary status is the greatest status whose audit gate is satisfied:

$$\sigma_\Pi(o) = \max_{\preceq}\{p \in \mathcal{P} : m_p(o, \Pi) = 1\}.$$

The predicate $B_D(o) = 1$ only when the review gate records completed review, authorized adoption, human authorization, jurisdiction assumptions, a contestability channel, adoption reasons and contestation record. Failure flags then apply a cap operator $\kappa_F : \mathcal{P} \to \mathcal{P}$:

$$PS_\Pi(o) = \min_{\preceq}\{c(R(o)), \kappa_{F(o)}(\sigma_\Pi(o))\}.$$

Warning flags leave the status unchanged while recording a defect. Downgrade and suspension flags cap external status at reference information unless the missing artefact is restored. Withdrawal flags cap status at $p_0$.

**Proposition 1 (Chain dominance and non-substitution).** For any policy $\Pi$ and output $o$, procedural status is fixed by the legal-material chain: $PS_\Pi(o) = \min_{\preceq}\{c(R(o)), \kappa_{F(o)}(\sigma_\Pi(o))\}$. If any gate required for $p_3$ is false, then $PS_\Pi(o) \prec p_3$ regardless of total audit score, upstream recall or authority coverage; if $B_D(o) = 0$ or $T < 2$ or $L < 2$, then $PS_\Pi(o) \prec p_4$; and if $F(o)$ is a withdrawal flag, then $PS_\Pi(o) = p_0$. Therefore, on any validation domain containing a gate-contrast pair $(o^+, o^-)$ with the same audit score vector, upstream metrics and system role but different value for a mandatory predicate or derived cap predicate, no score-only or retrieval-metric-only rule can reproduce $PS_\Pi$. A rule that omits any predicate in $\{B_A, B_E, B_M, B_P, B_C, B_J, B_T, B_D, B_F\}$ must assign the pair the same status, while the protocol assigns different maximum statuses.

This follows from $m_{p_3}$, $m_{p_4}$ and $\kappa_F$: screening requires $B_A \wedge B_E \wedge B_M \wedge B_P \wedge B_C \wedge B_J \wedge B_T \wedge B_F$; decision support adds $\sum \mathcal{A}(o) \geq \tau_D$, $T = L = 2$ and $B_D(o)$; role and failure caps apply after the score candidate. The implementation tests the proposition through formal invariants for gated monotonicity, gate non-substitutability, evidence-packet necessity, claim-anchor necessity, authority-gate necessity, counter-material necessity, contestability-channel necessity, decision-adoption necessity, role-cap dominance, failure-cap absorption and metric non-equivalence.

Three consequences follow. *Separation*: perfect authority coverage with non-procedural source tags or missing output-source links remains at $PS_{\Pi}(o) \preceq p_1$ when external screening is claimed. *Screening necessity*: an output cannot reach $p_3$ unless each audit dimension, evidence packet, material claim anchor, authority gate, counter-material gate, jurisdiction profile, source tag and contestability-channel route is reviewable. *Adoption necessity*: an output cannot reach $p_4$ unless $T = L = 2$ and $B_D(o) = 1$, even if it already qualifies as screening. These protocol properties are tested below on public, model-generated, repaired and adversarial packets.

## 5.2 Audit dimensions

The protocol evaluates the legal-output evidence packet rather than model parameters. A legal procedure may not need source code to decide whether a ranked case list can be contested. It needs enough information to reconstruct the legal-material chain and to test whether a relevant authority was omitted, misranked, outdated or mischaracterized.

## 5.3 Status thresholds

The audit vector determines the maximum status available to an output. A high aggregate score does not compensate for failure of a necessary gate. The model therefore uses both minimum gates and total-score bands. Table 3 states the default allocation rules.

The mapping function $g$ is therefore explicit rather than discretionary. First, assign the highest status whose gates are satisfied:

$$g_0(o) = \begin{cases} p_4, & m_{p_4}(o) = 1, \\ p_3, & m_{p_3}(o) = 1, \\ p_2, & m_{p_2}(o) = 1, \\ p_1, & m_{p_1}(o) = 1, \\ p_0, & \text{otherwise.} \end{cases}$$

Second, apply system-role and failure caps. A disclosed assistance tool cannot exceed professional support even if the audit vector would otherwise satisfy normative screening; an auditable procedural tool cannot become a decision-support reason without authorized adoption. Withdrawal-level failures, such as unreconstructable sources, materially false generated summaries or irreversible external action without human au-

Table 2: Audit dimensions for case recommendation outputs

| Dimension | Score 0 | Score 1 | Score 2 |
| --- | --- | --- | --- |
| $S$: **Source and corpus** | Sources or corpus cannot be reconstructed. | Individual source links exist, but corpus scope or update status is incomplete. | Source links, corpus manifest, inclusion and exclusion rules, update date and document versions are recorded. |
| $Q$: **Query and version** | Query, filters or output version are unavailable. | Some query or version information is retained. | Query, filters, issue labels, source-collection version, workflow or interface version and output identifier are reconstructable. |
| $H$: **Authority hierarchy** | Output does not distinguish legal status. | Output marks broad categories such as binding or persuasive. | Output records jurisdiction, court level, legal force, validity, later treatment and relation to governing norms. |
| $K$: **Ranking and counter-materials** | Ranking is opaque and contrary materials are not surfaced. | Broad ranking factors or some contrary materials are shown. | Ranking/re-ranking factors, top-$k$ set, omitted-material checks and counter-authority recall are reviewable. |
| $T$: **Contestability** | Affected persons cannot inspect or challenge the output. | A professional or internal reviewer can challenge the output. | Parties or reviewers can inspect, challenge, supplement and obtain a recorded response. |
| $L$: **Adoption and responsibility** | No adoption path or responsible actor is recorded. | Output and user action are logged. | Output ID, query, result list, adopted and rejected items, reasons, timestamp and responsible reviewer are logged. |

Table 3: Status allocation rules

| **Candidate status** | Necessary gates | Additional condition |
|---|---|---|
| **Reference information** | $S \geq 1$ and $Q \geq 1$ | Output remains internal or user verifies sources before external reliance. |
| **Professional support output** | $S, Q, L \geq 1$ | Professional user accepts verification responsibility and can explain limits. |
| **Normative material screening output** | $S, Q, H, K, T, L \geq 1$ and all screening gates satisfied | Total score $\sum \mathcal{A}(o) \geq 9$; authority set, evidence packet, material claim anchors, source tags, counter-materials, jurisdiction assumptions, review gate and contestability channel are present. |
| **Decision-support reason** | Normative-screening gates, $T = L = 2$ and $B_D = 1$ | Total score $\sum \mathcal{A}(o) \geq 10$ plus completed review, human authorization, adoption reasons and recorded contestation. |
| **No external legal effect** | Withdrawal-level failure, unaccountable external disposition, or rights-affecting use without authorization, review, contestability or accountability | Output is barred from external legal effect. |

thorization, set $PS(o)$ to no external legal effect. Suspension or downgrade flags, such as high-authority omission, invalid-authority recommendation, counter-material suppression, missing source-attribution tags, absent jurisdiction assumptions, review-gate failure, ranking drift or contestation failure, cap the output at reference status until the missing artefact is restored. Thus $g$ combines the gate function $g_0$ with role and failure caps. If $K = 0$, the output cannot be treated as normative material screening because the ranking and counter-material treatment cannot be reviewed. If $T < 2$, $L < 2$ or $B_D(o) = 0$, the output cannot be a decision-support reason, even if retrieval accuracy is high. If the system purports to trigger an external disposition without legal authorization, review, contestability or accountability, the output has no external legal effect.

Operationally, an evaluator applies the model in six steps: identify the candidate procedural status claimed for the output; freeze the relevant issue, query, source collection, workflow version and output identifier; score each component of $\mathcal{A}(o)$ using the audit artefacts in Table 2; apply the necessary gates in Table 3; assign the highest available status or record the missing gate; and repeat the audit after material corpus, workflow or interface changes. This makes the framework usable as a deployment checklist, post-use audit and benchmark extension.

The thresholds use non-substitutable gates. The 0–2 scale is ordinal: 0 means absent, 1 means incomplete or internal, and 2 means reviewable by the relevant professional or procedural actor. The default $\tau_N = 9$ permits one limited weakness while still requiring every gate; decision-support status requires $T = L = 2$ because external reasons require both contestation and adoption records. The full validation suite re-evaluates all 264 packets under thresholds 8–12. Thresholds 8–10 produced identical status allocations; stricter thresholds generated demotions, and no promotions occurred. Table 4 reports the full-packet sensitivity results.

Table 4: Full-packet sensitivity of status allocation to threshold policy

| $\tau_N$ | $\tau_D$ | Flips | Demotions | Status distribution |
|---|---|---|---|---|
| 8 | 10 | 0 | 0 | decision 12; screening 51; professional 13; reference 156; no effect 32 |
| 9 | 10 | 0 | 0 | decision 12; screening 51; professional 13; reference 156; no effect 32 |
| 10 | 11 | 0 | 0 | decision 12; screening 51; professional 13; reference 156; no effect 32 |
| 11 | 12 | 2 | 2 | decision 12; screening 49; professional 15; reference 156; no effect 32 |
| 12 | 13 | 58 | 58 | decision 0; screening 17; professional 59; reference 156; no effect 32 |

## 5.4 Jurisdictional parameterization

The audit dimensions are stable, but the materials satisfying $H$ and $K$ are local. For each setting $j$, evaluators instantiate authority hierarchy as $H_j$ and counter-material handling as $K_j$ before scoring. A high-status output must also carry jurisdiction assumptions that support the declared profile; an otherwise complete packet is downgraded when the profile is absent or mismatched. Table 5 gives the profile mapping.

Table 5: Jurisdictional profiles for $H_j$ and $K_j$

| **Setting** | $H_j$: authority hierarchy | $K_j$: counter-materials and limiting materials |
|---|---|---|
| **Common law** | Binding precedent, persuasive precedent, statutes, subsequent treatment, overruled or superseded authorities. | Distinguishing cases, contrary lines, limiting subsequent treatment, minority lines and overruled authorities. |
| **Civil law** | Constitutional norms, code provisions, special statutes, regulations, high-court decisions, lower-court decisions and doctrinal commentary. | Competing statutory interpretations, systematic conflicts, teleological limitations, contrary doctrinal views and later amendments. |
| **Admin. adjudication** | Statutes, binding regulations, agency rules, official guidance, adjudicative decisions and non-binding policy documents. | Conflicting agency interpretations, withdrawn guidance, higher-level statutory constraints and contrary adjudicative decisions. |
| **Arbitration** | Contract text, chosen law, mandatory law, institutional rules, prior awards and trade usage. | Contrary awards, mandatory-law limits, distinguishing clauses, procedural fairness objections and public-policy exceptions. |
| **ODR** | Platform rules, consumer statutes, terms of service, regulatory guidance and prior platform decisions. | Consumer-protection overrides, exception policies, appeal decisions and human-review overrides. |

## 5.5 Contestability and audit responsibility

Contestability requires a practical opportunity to inspect selected materials, identify missing or contrary materials, submit alternatives, request human review and receive a recorded response when the output is adopted. Adoption logging is distinct: it records whether, why and by whom the output was taken up after contestation was possible. Audit responsibility assigns each artefact to the actor that controls it: developers for capability and safety claims, deployers for corpus and interface records, professional users for issue framing and verification, and authorized adopters for decision uptake.

# 6 Operationalizing the Framework for Case Recommendation

AI-based case recommendation can be produced by many upstream architectures, but the audit framework does not depend on their internal mechanics. The legally relevant object is the output evidence packet: the source collection, versions, legal materials, output units, source locators, authority labels, counter-material links and adoption records that make the output reconstructable. Failures in these artefacts create different legal consequences. A stale source collection creates source and temporal risk. An ordering rule that privileges surface similarity may miss legally decisive distinctions. A generated summary may misstate holdings or flatten procedural posture. A user interface may display confirming cases while hiding contrary authorities.

The evaluation protocol should therefore separate upstream performance metrics from procedural checks. Precision, recall, F1, MRR and related metrics may help evaluate an upstream retrieval or recommendation component. Procedural evaluation asks whether the output has preserved the legal hierarchy of authorities, displayed counter-materials, distinguished binding and persuasive materials, recorded update status, and produced a contestable adoption trail. These are evaluation requirements for procedural status.

The procedural metrics can be stated directly. *Authority coverage* asks whether known high-authority materials for the issue appear in the output set. *Counter-authority recall* asks whether legally relevant contrary or limiting materials are surfaced. *Hierarchy accuracy* asks whether the output correctly labels binding, precedential, persuasive, overruled or superseded materials. *Version traceability* asks whether the source collection and upstream run state that produced the output can be reconstructed. *Procedural-source coverage* asks whether source links are bound to official, tool-verified or otherwise procedurally usable source tags rather than just named. *Adoption-log completeness* asks whether later human use of the output is observable. These metrics convert upstream benchmark results into procedural-use evidence. Table 6 turns these distinctions into output-qualification cases.

## 6.1 Baseline testing

Baseline testing should cover both technical and legal dimensions. A case recommendation system should be tested against known authority sets, especially high-authority or binding materials. Missing a binding, precedential, designated or otherwise high-authority material is different from missing one marginally similar low-level case. The former may trigger withdrawal from procedural use even if aggregate retrieval metrics remain acceptable.

Table 6: Decision matrix for legal AI output qualification

| **Output use** | Minimum verifiability | Minimum contestability | Allowed procedural status |
|---|---|---|---|
| **Internal legal search** | Source and version record; user can inspect retrieved materials | Internal user review | Reference information |
| **Professional advice support** | Source, query, output and limitations recorded | Client or supervising professional can request explanation when output materially shaped advice | Professional support output |
| **Court or tribunal case recommendation** | Corpus scope, update date, authority hierarchy, ranking factors, counter-authority handling and version record | Parties can inspect, challenge and supplement recommended materials | Normative material screening output |
| **Decision reasoning support** | Full adoption path, human confirmation, contrary-material treatment and audit trail | Parties can contest use; decision-maker gives reasons for adoption or rejection | Decision-support reason |
| **Unaccountable external disposition** | Missing legal authorization, review, contestability or accountability chain | Not satisfied | No external legal effect |

Baseline testing should also record database coverage, update intervals, jurisdictional boundaries, language coverage, case hierarchy, citation treatment and exclusion rules. A system trained or indexed on a narrow corpus should not be evaluated as if it were a general legal research system. A system that excludes unpublished decisions, local court decisions or outdated but historically relevant authorities should make that exclusion visible to the user.

## 6.2 Counter-material presentation

Case recommendation is procedurally dangerous when it confirms one line of reasoning without exposing plausible counter-materials. A useful legal recommendation system should therefore include counter-authority handling as a first-order evaluation requirement. The goal is to preserve the adversarial and reason-giving structure of legal reasoning by ensuring that contrary cases, limiting cases, overruled authorities and minority lines are visible when they are legally relevant.

For audit purposes, counter-authority recall (CAR) can be measured against a manually curated issue set:

$$CAR = \frac{|\text{retrieved counter-authorities} \cap \text{known counter-authorities}|}{|\text{known counter-authorities}|}.$$

The denominator is not the universe of all contrary law. It is the set of contrary or limiting materials identified for a defined issue by the relevant court, professional reviewer or benchmark curator. This makes the measure imperfect but usable.

This requirement is particularly important for generated legal outputs. A ranked list may visibly display multiple authorities. A generated summary may compress them into a single narrative and thereby hide disagreement. Evaluation should test whether the output cites supporting materials accurately, distinguishes holding from dicta, distinguishes binding from persuasive authorities, and signals unresolved conflict instead of hallucinating consensus. Empirical work on legal hallucinations shows why source verification cannot be left to model fluency (Dahl et al., 2024).

## 6.3 Output evidence packets

The audit layer proposed here sits above any particular search, retrieval, database, generation, agent or manual review architecture. Procedural status attaches to an *output evidence packet*: a provider-agnostic record that connects the legal propositions or recommended materials in an output to source identifiers, versions, locators and review status.

A minimally reviewable evidence packet records the issue, source collection, source version, output identifier, output units, linked source identifiers, locators, source-attribution tags, authority labels, counter-material links, human review gate, adoption record and objection record. Unlike a model card or dataset sheet, the documented unit is the procedurally relevant legal output (Gebru et al., 2021; Mitchell et al., 2019). Whether the upstream system is search, database retrieval, generation, agentic workflow or manual review, the audit question is the same: can the legal-output chain be reconstructed, source-tagged, reviewed and contested?

Four structural packet metrics make this requirement operational. For compact reporting below, the article abbreviates structural evidence fidelity as EF, evidence coverage as EC, source-tag coverage as STC and procedural-source coverage as PSC. *Structural evidence fidelity* is the proportion of output-source links marked as supporting the output unit to which they are attached:

$$EF = \frac{|\text{output-source links that support the unit}|}{|\text{all output-source links}|}.$$

*Evidence coverage* is the proportion of legally material output units that have at least one reconstructable source locator:

$$EC = \frac{|\text{output units with source-locator support}|}{|\text{all legally material output units}|}.$$

*Source-tag coverage* measures output-source links carrying a verification-status tag:

$$STC = \frac{|\text{output-source links with source-attribution tags}|}{|\text{all output-source links}|}.$$

*Procedural-source coverage* is stricter:

$$PSC = \frac{|\text{links tagged official, tool-verified, user-verified or settled}|}{|\text{all output-source links}|}.$$

The tag may indicate whether the source was verified by a connected legal research tool, drawn from public metadata, supplied by the user, treated as settled background, or still requires pinpoint verification. These packet-integrity metrics make structural reviewability explicit and route legal merits judgment to professional or institutional review. For external normative screening, a "needs verification" tag records unfinished source work. Low $EF$, $EC$, $STC$ or $PSC$ fails procedural reconstructability even if an upstream system found relevant documents.

## 6.4 Review and adoption gates

Source reconstruction does not by itself authorize legal reliance. The gate record should state whether professional review is required and completed, what jurisdiction assumptions were used, which reliance level applies, whether any irreversible external action has human authorization, and what the human actor did with the output. An accurate draft remains below decision-support status until an authorized actor reviews it, adopts it and records reasons.

The minimum adoption-log schema is: output identifier, time, user role, issue, filters, source-collection version, workflow version, displayed material set, displayed counter-materials, source-attribution tags, review-gate status, adopted item, rejected item, human reason, objection or supplement submitted, response to objection and final responsibility node. A system that cannot export these fields should remain below decision-support status.

# 7 Downgrade and Withdrawal Conditions

Procedural status is dynamic. Corpus changes, workflow updates, ranking drift, source-attribution gaps, review-gate failures or contestation failures can trigger failure caps after an output or system was initially approved. Table 7 operationalizes those caps: warnings correct isolated defects, downgrades cap status until a missing artefact is restored, suspensions block external use for an issue class, and withdrawals remove external use when the legal-material chain cannot be reconstructed or an irreversible action lacks human authorization.

# 8 Implementation and Evaluation

The rules above are implemented as an executable audit harness and proof-carrying allocation layer. The implementation uses local JSON scenario files as the unit of evidence and produces Markdown, JSON and certificate artefacts. The companion artifact supplies the harness, scenario files, source snapshots, run artefacts and replication commands.

The harness implements the mapping in Section 4 directly. Each scenario declares a claimed status, role evidence, audit vector, jurisdiction profile, upstream metrics and evidence packet. Authority sets, review gates and counter-material declarations are mandatory for screening and decision-support claims; missing fields trigger caps. The runner computes status without reading the expected outcome and emits metrics, dispositions and proof certificates.

## 8.1 Evaluation design

The evaluation is not a retrieval leaderboard. It is a falsification design for a procedural-status predicate. The matrix asks four validation questions. First, can the protocol allocate status correctly when the legal-material chain is controlled? Second, does the rule survive public outputs, raw model outputs, source-supported

Table 7: Downgrade severity levels

| Severity | Trigger | Procedural consequence |
|---|---|---|
| **Warning** | Isolated error with reconstructable source, query and adoption record. | Correct the output and monitor the issue class. |
| **Downgrade** | Missing authority hierarchy, incomplete counter-material display, missing source tags, incomplete review gate or weak adoption logging. | Limit output to reference or professional support status until the missing artefact is restored. |
| **Suspension** | Repeated omission of high-authority materials, invalid authority recommendation or unexplained ranking drift in an issue class. | Suspend external procedural use for that domain or data source. |
| **Withdrawal** | Source, corpus, ranking or adoption chain cannot be reconstructed, the system generates materially false legal summaries, or an irreversible action lacks human authorization. | Withdraw from external legal use; retain only internal testing or research use. |

repairs and withheld packets? Third, can simpler substitute theories reproduce the allocation? Fourth, is the allocation stable under policy variation, architecture and model labels, challenge events, certificate replay, tampering and coding uncertainty? Table 8 summarizes the validation questions. This organization separates validation units from generated operations, so the scale of the matrix supports the claim rather than replacing it.

Table 8: Evaluation design

| **Question** | Target threat | Evidence layers | Required result |
|---|---|---|---|
| **VQ1** | The status function is decorative or under-specified. | Stress scenarios, construct-operationalization coverage, formal invariants, status-lattice exhaustion, policy-constants replay and proof-carrying certificates. | Gates, role caps, failure caps, construct links and proof obligations reproduce the stated status rule. |
| **VQ2** | The rule works only on constructed examples. | Public metadata, public-system outputs, endpoint-matched public retrieval, issue-specific public packets, holdouts, raw model outputs and source-supported repairs. | Public and model outputs are classified by visible artefacts, not by fluency, provider or raw retrieval success. |
| **VQ3** | A simpler rule explains the results. | Metric separation, baseline comparison, issue-family holdout, multi-axis holdout, policy-family robustness, gate-contrast witnesses and policy mutation. | Retrieval metrics, total score, source labels, authority coverage, review labels and model identity fail to reproduce the full allocation. |
| **VQ4** | Results are brittle to coding, architecture, contestation or certificate manipulation. | Source-chain attacks, contestation challenges, workflow portability, model-identity invariance, claim-anchor analysis, review-provenance analysis, certificate tamper-resistance, recoding, uncertainty and blind coding. | High status survives only when the legal-material chain remains reconstructable, challengeable, replayable and source-anchored. |

The complete evaluation matrix adjudicates one rule: procedural status follows the legal-material chain. It contains 264 scenario packets with 697 embedded records or outputs and 8,023,761 validation operations across public-output, model-output, cross-engine, architecture-portability, identity-substitution, issue-family holdout, multi-axis holdout, policy-family robustness, invariant, status-lattice, adversarial, ablation, mutation, review-provenance, claim-anchor, replay, certificate, query, ranking, construct-operationalization coverage, threat-model coverage and coding-robustness layers. The main text should be read as an evidence architecture rather than as a catalogue of runs. Public-output and holdout layers test whether visible legal records can be procedurally classified. Model-output and repair layers test whether fluent authority recommendations change status when source support, counter-material salience and adoption gates change. Cross-engine, workflow-portability and model-identity layers test whether runtime architecture and provider identity add status once the legal-material chain is fixed. Metric, baseline, holdout, policy-family, invariant and lattice layers test six rival theories–performance sufficiency, source-label sufficiency, authority-material sufficiency, review-label sufficiency, score sufficiency and model-identity sufficiency. Attack, challenge, claim-anchor, replay, mutation and tamper layers test whether any broken link destroys qualification.

The full claim-by-claim map is preserved in the generated validation report. Table 9 is the reviewer roadmap for interpreting it in the manuscript; Tables 10, 11, 12 and 13 report the underlying layer counts and headline results.

The target variable is procedural allocation. A positive packet is one that the full protocol qualifies for normative screening or decision support after applying score gates, evidence-packet gates, role caps and failure caps. In the tables, FP and FN mean false positives (FP) and false negatives (FN). The generated report emits a substitute-theory falsification table. Recall-threshold performance produced 153 scenario false positives; source-label sufficiency produced 72 scenario false positives and 6,552 lattice false positives; authority-material sufficiency produced 150 scenario false positives, 2 scenario false negatives and 672 lattice false positives; review-label sufficiency produced 149 scenario false positives and 19,800 lattice false positives; score sufficiency produced 183 scenario false positives and 49,752 lattice false positives; model-identity sufficiency produced 1,005 identity-substitution false positives. The full protocol produced zero false positives and zero false negatives in the same comparison.

The status-lattice layer is a finite-state characterization of high-status claim attempts. It fixes the output claim at decision-support level and enumerates 466,560 score-role-gate states, 3,499,200 cover edges and 3,732,480 substitute-rule predictions. Only 168 states reached high status and only 11 reached decision-support status. All 1019/1019 high-status necessity checks and 1019/1019 gate-ablation drops passed. The extra lattice gate is claim-anchor completeness: a source link must point to a material proposition, not merely to a packet-level source list. Cover-edge demotions are diagnostics: partial gate additions create incomplete external-use chains, and incomplete chains are supposed to drop. The best partial substitute still admitted 672 false posi-

Table 9: Evidence architecture for reading the validation matrix

| **Claim** | Main evidence layer | What would falsify the claim |
|---|---|---|
| **The protocol is executable** | Formal invariant verification, status-lattice exhaustion and policy-constants replay | Gates, caps or downgrade rules cannot reproduce the stated status function. |
| **Retrieval success is not procedural qualification** | Public retrieval, raw model-output audit, metric separation and baseline comparison | High authority coverage, recall, F1 or total score upgrades an output without the legal-material chain. |
| **Policy choices do not drive the core result** | Policy-family robustness | Admissible threshold, salience, source-tag, role-cap or failure-severity variants promote below-screening packets or let simplified substitutes reproduce the protocol. |
| **Status is architecture-portable and deployment-role bounded** | Workflow portability analysis | Runtime architecture labels change status, entitlement profiles exceed role caps, decision status appears without a screening-capable chain, or unaccountable external use survives. |
| **Model identity is not procedural status** | Cross-engine outputs and model-identity invariance | Provider, model or engine labels change status while the legal-material chain is unchanged. |
| **Qualified outputs depend on their gates** | Gate ablation, gate-contrast witnesses, source-chain attacks and claim-anchor analysis | Removing source, authority, counter-material, review, contestability, jurisdiction or adoption predicates leaves high status intact. |
| **Status allocation is replayable and tamper-resistant** | Status-certificate replay, certificate tamper-resistance and policy-constants replay | A certificate, policy replay or proof obligation cannot be reproduced, or a tampered field survives replay. |
| **Core constructs and validity threats are covered** | Construct-operationalization coverage and threat-model coverage | A construct or validity threat lacks independent committed evidence layers and evidence roles. |

Table 10: Full evaluation matrix

| Layer | Units | Rule pass | What it checks |
|---|---|---|---|
| **Stress scenarios** | 10 | 10/10 | Status gates, failure caps, downgrade and withdrawal behavior. |
| **Public metadata** | 120 records in 6 files | 6/6 | Source reconstruction across six public legal-record sources. |
| **Public-system outputs** | 60 records in 6 files | 6/6 | Real upstream legal-output reconstruction and professional-support caps. |
| **Issue-specific public output/source packets** | 19 records in 3 files | 3/3 | Public issue-search outputs and one source-bound public-source packet tested against high-authority and counter-material requirements. |
| **Endpoint-matched public retrieval benchmark** | 169 records in 22 files | 22/22 | Public case-law or known-item outputs tested against authority sets that the endpoint can return; mixed authority gaps are recorded separately. |
| **Out-of-sample holdout validation** | 126 items in 24 files | 24/24 | Withheld public retrieval packets, raw model-output packets and source-bound repairs tested after the scoring policy was frozen. |
| **Raw model outputs** | 10 | 10/10 | High authority coverage without source binding remains procedurally capped. |
| **Source-supported model-output repairs** | 10 | 10/10 | Same model outputs qualify after a validator checks manifest membership, locators, hashed source-support excerpts, source tags and issue-set coverage. |
| **Cross-engine raw model outputs** | 9 | 9/9 | Three model engines across three issue families remain capped without source-bound legal-material chains. |
| **Cross-engine source-supported repairs** | 9 | 9/9 | The same cross-engine outputs qualify after source-support validation, regardless of model identity. |
| **Model-output evidence ladder** | 70 | 70/70 | Same model outputs move across reference, professional-support, screening, decision-support and no-effect status when evidence and adoption gates change. |
| **Adversarial source-support repairs** | 60 | 60/60 | Negative controls test locator, claim, contradiction, source, link and counter-material attacks against the repair validator. |
| **Public source-text anchors** | 30 checks | 30/30 | Support terms are checked against extracted public source text snapshots, reducing manifest-only validation. |
| **Model-output transcript anchors** | 50 checks | 50/50 | Raw model-output scenario locators are checked against the committed transcript sections. |
| **Cross-engine transcript anchors** | 36 checks | 36/36 | Cross-engine raw-output locators are checked against committed transcript sections. |
| **Formal invariant verification** | 51,646 checks | 51,646/51,646 | Exhaustive generated states test monotonicity, authority-set necessity, evidence-packet necessity, claim-anchor necessity, counter-material necessity, contestability-channel necessity, adoption gates, role caps, failure caps and metric non-equivalence. |
| **Status-lattice exhaustion** | 466,560 claim-attempt states / 3,499,200 edges | 1019/1019; 1019/1019 | Exhausts high-status claim attempts; high-status necessity and gate-ablation drops must hold. |

Table 11: Full evaluation matrix: diagnostics and robustness I

| Layer | Units | Rule pass | What it checks |
|---|---|---|---|
| **Metric separation** | 219 evaluations | Recall-threshold precision 0.29; reference gate FP 0 | Tests whether upstream precision, recall and F1 reproduce protocol-defined procedural qualification. |
| **Metric statistical robustness** | 2,000 resamples | Bootstrap CI for recall-threshold precision 0.23–0.35; permutation $p = 0.354$ | Tests whether the retrieval/status separation is stable under resampling and label permutation. |
| **Baseline rule comparison** | 3,252 predictions | Best simplified false positives 38; reference rule false positives 0 | Tests whether recall, F1, total-score, source-bound, authority-material or review-gate substitutes can reproduce the full audit allocation. |
| **Issue-family generalization** | 3,014 holdout predictions | Full protocol FP/FN 0/0; best trained simplified FP 38 | Holds out each of 5 issue families; simplified substitute rules selected on the remaining families err in 5/5 held-out folds. |
| **Multi-axis holdout generalization** | 19,235 holdout / 91,333 training predictions | Full protocol FP/FN 0/0; best trained simplified FP 510 | Holds out issue family, validation suite, system role, jurisdiction profile, claimed status and output-origin strata. |
| **Policy-family robustness** | 42,192 evaluations | 0 high-status promotions; 12/12 variants with simplified-rule errors | Varies thresholds, rank-salience windows, procedural source tags, role caps and failure severity; best simplified rules produce 449 cumulative false positives while full protocol FP/FN remains 0/0. |
| **Qualified-output gate ablations** | 390 ablations | 390/390 | Removes procedural gates from all qualified packets and checks that status falls below the corresponding level. |
| **Gate-contrast witness pairs** | 390 pairs | 390/390 | Preserves audit scores, upstream retrieval metrics and system role while changing one mandatory predicate; every witness separates procedural status. |
| **Source-chain attacks** | 1,953 variants | 1,953/1,953 | Applies every nonempty combination of five source-chain attacks across qualified packets; all variants lose high status through downgrade, suspension or withdrawal despite high audit scores and high upstream recall. |
| **Dynamic contestation challenges** | 315 variants | 315/315 | Valid challenges block high status in 252/252 cases; unsupported challenge controls preserve high status in 63/63 cases. |

tives; the full screening predicate admitted 0 false positives and 0 false negatives. This is the formal core of the article: procedural status is a gated legal-material relation, not a scalar score.

Across ten stress-test scenarios, all rule outcomes were reproduced. The mean audit score was 8.30 and mean upstream recall was 0.84. Three scenarios had upstream recall above 0.80 but were procedurally blocked or downgraded. A high-recall output is unfit for external legal use when it omits a high-authority material, suppresses a relevant limiting material, recommends invalid authority, lacks output-source evidence or prevents affected persons from contesting the output.

## 8.2 Public legal-record reconstruction

The harness was also run on public legal-record snapshots. One layer sampled 20 public metadata records from each of six legal systems. A second layer froze the top ten records visible in six public legal retrieval or listing streams. These layers test reconstruction: whether visible records can be frozen, source-tagged, linked to locators and assigned a procedural cap by the audit layer. The protocol assigned all six files to professional support: public record visibility establishes source reconstructability, while normative screening requires an issue-defined authority chain, counter-material completeness and a contestation route.

Table 12: Full evaluation matrix: diagnostics and robustness II

| Layer | Units | Rule pass | What it checks |
|---|---|---|---|
| **Metamorphic policy tests** | 1,233 transformations | 1,233/1,233 | Claim escalation, metric inflation, role-cap demotion, source-tag mutation, review-gate removal, score-and-role inflation without adoption, and benign-source augmentation satisfy the expected status-function relations. |
| **Policy mutation analysis** | 3,111 evaluations | 15/15 killed | Gate-removal and status-conferring policy mutants are killed when they remove required gates, ignore caps, or treat metrics, source binding, authority coverage, review labels, total score or model identity as status-conferring. |
| **Review-provenance analysis** | 627 evaluations | 627/627 | Review/adoption labels cannot supply a missing legal-material chain; missing contestability channel, incomplete review, missing jurisdiction assumptions, missing authorization, missing adoption reasons and missing contestation record demote high-status claims. |
| **Claim-anchor analysis** | 1,080 mutations | 1,080/1,080 | Removes material proposition text, link-to-claim bindings, support attestations and locators from qualified packets; high status survives only when each claim remains source-anchored. |
| **Workflow portability analysis** | 2,904 mutations | 2,904/2,904 | Five runtime-architecture labels leave status invariant; four entitlement profiles obey role caps; decision support appears only when the procedural profile has a screening-capable chain; unaccountable external use is barred. |
| **Model-identity invariance** | 1,320 substitutions | 1,320/1,320; 0 status changes | Replaces provider, model and engine identity labels across all 264 packets using five identity profiles; status and disposition remain fixed. |
| **Query-perturbation stability** | 30 variants / 5 groups | 5/5 status-stable groups | Compares issue-equivalent public-search queries; authority coverage is unstable in 3/5 groups, returned-record sets in 4/5 groups and top results in 4/5 groups. |
| **Query-portfolio frontier** | 315 portfolios / 5 groups | 0/315 qualified | Enumerates all non-empty query portfolios; 56/315 recover full high-authority coverage, but 0/315 recover full counter-material or screening-material coverage. |
| **Construct-operationalization coverage** | 10 constructs / 40 checks | 40/40 | Maps the paper's core constructs to committed evidence layers. |
| **Threat-model coverage** | 8 threats / 32 checks | 32/32 | Maps construct, internal, source-chain, external, architecture, policy, provenance and annotation validity threats to committed evidence layers. |
| **Blocked-claim repair frontiers** | 5,097 counterfactuals | 193/193 | Computes minimal artifact-gate families needed to upgrade blocked normative-screening or decision-support claims. |
| **Jurisdiction-profile mutations** | 440 checks | 189/189 | Valid generic profile assumptions preserve status, while missing or mismatched profile assumptions downgrade qualified packets. |
| **Ranking-visibility diagnostics** | 956 window checks / 85 counterfactuals | 85/85 | Computes counter-material visibility across rank windows; rank-order counterfactuals preserve authority coverage and counter-material recall while activating ranking-drift caps. |
| **Proof-carrying status certificates** | 6,072 checks / 4,488 obligations | 6,072/6,072; 4,488/4,488 | Replays proof-carrying status certificates for all 264 packets from scenario identity, scenario hash, policy hash, policy body, jurisdiction profile, score gate, role cap, missing gates, failure cap, metrics, claim support, proof obligations and derivation hash. |
| **Certificate tamper-resistance** | 5,811 cases | 5,811/5,811 | Mutates certificate identities, hashes, policy bodies, scores, roles, gates, status fields, caps, metric bundles, proof obligations and certificate-set structure. |
| **Policy-constants replay** | 4,752 checks | 4,752/4,752 | Recomputes score candidates, role caps, protected reliance gates, missing gates, failure caps, metrics and final status in a separate script parameterized by JSON policy constants without importing the harness model. |

Table 13: Full evaluation matrix: coding and sensitivity

| Layer | Units | Rule pass | What it checks |
|---|---|---|---|
| **Mixed-authority public source-screening packets** | 5 | 5/5 | Source-bound statute, case and source packets can qualify as normative screening. |
| **Issue ablations** | 20 | 20/20 | High-authority omissions, counter-material suppression, unverified source tags and missing adoption gates trigger caps. |
| **Annotation robustness** | 528 recodings | 262/264 stable | Strict and lenient recoding policies test status stability across all packets. |
| **Annotation uncertainty** | 66,000 perturbations | 0.937 sample stability; 0.916 qualified high-status stability | Fixed-seed score-noise perturbations locate threshold-boundary cases and test coding-noise robustness. |
| **Score-blinded dual coding** | 240 packets, 2 passes | 0.99 coder kappa; 0.93 min dimension kappa; 0.87 min base exact | Coding passes score non-holdout packets and compare status, inter-coder dimensions, derived gates and base calibration. |
| **Threshold sensitivity** | 1320 evaluations | 0 flips at thresholds 8–10 | All 264 packets are re-evaluated under normative thresholds 8–12. |

## 8.3 Issue-specific public output/source audit

A stricter public layer then froze two issue-specific public search outputs and one source-bound public-source packet, converting their visible records into evidence packets. The U.S. packet used a CourtListener issue search for agency deference after *Loper Bright*; the English-law packet used a National Archives issue search for mesothelioma causation after *Fairchild*; and the EU packet used selected Legislation.gov.uk and CURIA public source pages for General Data Protection Regulation (GDPR) Article 15 access-right materials. The layer contains 19 public records across three files.

The source-bound EU public-source packet qualified as normative material screening because it included the relevant high-authority materials, limiting provisions, source locators and reviewable output-source links. The U.S. and English issue-search outputs were capped at reference information because the visible returned records omitted high-authority or counter-material records for the defined issue. The procedural-status result is sharper than a relevance score: public legal records can be source-reconstructable yet fail normative-screening qualification when the output does not preserve the issue-specific authority and counter-material chain.

## 8.4 Public retrieval benchmark

The public retrieval benchmark generalizes this separation across 22 public retrieval outputs scored against endpoint-compatible authority sets. The benchmark covers five issue families: U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, Canadian administrative-law standard of review after *Vavilov*, German/EU right-to-be-forgotten review and EU GDPR Article 15 access rights. It uses CourtListener, the National Archives, Supreme Court of Canada Lexum search, OpenLegalData and CURIA public search or list endpoints. The benchmark contains 169 returned records. The returned top-$k$ records are the committed outputs exposed by the public endpoints for the selected queries. Its gold sets are matched to endpoint scope; mixed statute/case screening is tested in the source-bound issue packets.

All 22 outputs were capped at reference information. Mean high-authority recall was 0.23 and mean counter-material recall was 0.00. Source-reconstructable public search output is therefore not the same as normative material screening. A system can expose real cases, stable URLs and rank order while still failing the procedural requirement that high-authority and limiting materials for a defined legal issue be made visible together.

The query-perturbation diagnostic then grouped the 22 benchmark packets with eight withheld public-retrieval variants. Procedural status was stable in all five issue groups, but authority coverage, record sets and top results were unstable. Query expansion improved high-authority coverage in some portfolios, yet 0/315 portfolios recovered full counter-material coverage or qualified for screening. Broader query formulation improved visibility; it did not replace a counter-material protocol.

## 8.5 Out-of-sample holdout validation

After the scoring policy was frozen, the harness generated a separate holdout layer: eight withheld public-retrieval packets, eight raw model-output packets and eight source-bound repair packets. The holdout scenario files do not store expected statuses. This layer tests whether the rule learned from the main matrix still separates three conditions that matter for the paper's claim: public legal search output, unsupported model text and source-bound model-output repair.

The result reproduced the separation. All eight public-retrieval holdout packets and all eight raw model-output holdout packets were capped at reference information, including nine packets with high upstream performance. All eight source-bound repairs qualified as normative material screening. The audit layer assigned status from visible artefacts rather than from provider identity, output fluency or retrieval recall.

## 8.6 Raw model-output pilot

The harness was also run on ten raw legal-authority recommendations generated by a single upstream model configuration, Codex GPT-5.5 with extra-high reasoning. The prompts covered the same three issue families as the positive-control packets: U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, and EU GDPR Article 15 access rights. This closed-book transcript condition treats fluent model text as an upstream legal output and asks whether strong authority coverage can claim normative screening status.

It cannot. In the coded raw-output packets, all ten outputs identified the expected high-authority and limiting materials in the issue families, producing mean upstream recall of 1.00 and counter-authority recall of 1.00. Yet every claimed normative-screening output was downgraded to reference information because the citations were marked as requiring verification rather than bound to official sources, corpus manifests, pinpoint locators and a contestation pathway. Their source-tag coverage was 1.00, but procedural-source coverage was 0.00.

The repair intervention applies the opposite condition to the same authority sets. A validator checks each output-source link against an issue manifest, locator, hashed source-support excerpt, contradiction pattern, procedural source tag and issue-set coverage; a rank-salience repair moves retrieved limiting material into the review window. Transcript and public source-text verifiers anchored all 50 raw-output locators and all 30 public support items. All ten repaired variants qualified as normative material screening and none became decision-support reasons. This paired result is the paper's rule in executable form: authority coverage is not procedural status, but validator-backed source support and visible counter-material can make authority coverage procedurally screenable. Table 14 reports the raw-output issue-family summary.

The cross-engine layer generalizes the same contrast. The same authority-recommendation prompt and issue families were run across Codex GPT-5.5, Codex GPT-5.4-mini and Gemini 2.5 Pro. All nine raw outputs reached perfect issue-set recall and all nine remained reference information because their legal-material chains were not source-bound. The source-supported repair layer then held the issue families constant and supplied manifest-backed source links, locators, procedural source tags, counter-material salience and completed review gates. All nine repaired outputs qualified as normative material screening.

The model-identity invariance layer then closes the provider-label route. It substituted five identity profiles–frontier general model, legal-specialist vendor model, open-weight model, small fast model and undisclosed agentic stack–across all 264 packets while holding the output, scores, authority sets, evidence packets, review gates and failure flags fixed. Across 1,320 mutated packets, status changed zero times and disposition changed zero times. Model identity is not evidence. It is metadata. The legal-material chain, not the engine label, carries procedural status.

The workflow-portability layer ran 2,904 checks over all 264 packets. The 5 runtime-architecture labels–public search, retrieval-augmented generation (RAG), agentic workflow, commercial database and manual review–left status, disposition, failure flags, missing gates, scores and claim support unchanged in 1,320/1,320 mutations. The 4 deployment-entitlement profiles obeyed role caps in 1,056/1,056 checks; decision support depended on a screening-capable procedural chain in 264/264 checks; unaccountable external disposition was barred in 264/264 checks. This answers the architecture objection directly: the contribution is not a rule for one implementation stack, but an output-status model portable across legal AI workflows.

The evidence ladder then varies seven conditions over ten model-output variants. It produced 30 reference-information outputs, 10 professional-support outputs, 10 normative-screening outputs, 10 decision-support reasons and 10 outputs with no external legal effect. Generated legal content changes procedural status only when the legal-material chain, rank salience or adoption gate changes.

The same ten outputs were also converted into sixty adversarial repair variants. The attacks introduced locator mismatches, unsupported claims, contradiction-pattern claims, out-of-manifest sources, missing output-source links and counter-material omissions. All sixty adversarial variants were rejected despite mean upstream recall of 1.00: forty were capped at reference information, while twenty unsupported or contradictory source-support variants were withdrawn from external legal effect. The blocked-failure distribution across the full matrix shows why high-performing outputs failed: source-attribution gaps dominated, followed by summary distortion, counter-material suppression, authority omission and smaller numbers of contestability and invalid-authority failures.

Table 14: Raw model-output pilot

| **Issue family** | N | Mean score | Upstream recall | PSC | Allowed status |
|---|---|---|---|---|---|
| **U.S. agency deference after *Loper Bright*** | 3 | 9 | 1.00 | 0.00 | Reference information |
| **English mesothelioma causation after *Fairchild*** | 3 | 9 | 1.00 | 0.00 | Reference information |
| **EU GDPR Article 15 access rights** | 4 | 9 | 1.00 | 0.00 | Reference information |

To test mixed-authority normative material screening itself, the harness includes five public source-screening packets: U.S. administrative law after *Loper Bright*, English-law mesothelioma causation after *Fairchild*, EU GDPR Article 15 access-right interpretation, Canadian administrative-law standard of review after *Vavilov*, and German/EU fundamental-rights review after the *Right to be Forgotten* decisions. Each packet defines high-authority materials, limiting or counter-materials, invalidated or historical treatments, source tags and output-to-source links at the level of individual output units. These mixed statute/case/source construct tests ask whether a source-bound, issue-defined authority packet can obtain screening status under the protocol. All five qualified as normative material screening, while decision-support status remained reserved for authorized institutional adoption. Table 15 reports the five packets.

The ablation suite then removes one procedural condition at a time from those packets. Twenty ablations were generated: high-authority omission, counter-material suppression, unverified source tags and missing authorized-adoption gates for each issue family. Fifteen were capped at reference information through downgrade or suspension. The remaining five claimed decision-support status but were capped at normative screening because the packet lacked authorized institutional adoption. This is the crucial negative test: a packet that is otherwise strong on source links and authority coverage still loses status when the missing condition is procedural rather than semantic.

Table 15: Mixed-authority public source-screening packets

| Issue packet | Profile | Authority set | Metrics | Status |
|---|---|---|---|---|
| **U.S. agency deference after *Loper Bright*** | Common law | H=3; C=2; invalid=1 treatment | Auth.=1.00; CAR=1.00; EF=1.00 | Normative screening |
| **English mesothelioma causation after *Fairchild*** | Common law | H=3; C=2; invalid=1 treatment | Auth.=1.00; CAR=1.00; EF=1.00 | Normative screening |
| **EU GDPR Article 15 access rights** | Civil law/EU | H=3; C=2; invalid=2 treatments | Auth.=1.00; CAR=1.00; EF=1.00 | Normative screening |
| **Canada standard of review after *Vavilov*** | Common law | H=3; C=2; historical=1 treatment | Auth.=1.00; CAR=1.00; EF=1.00 | Normative screening |
| **Germany/EU right-to-be-forgotten review** | Civil law/EU | H=3; C=2; cross-system=2 treatments | Auth.=1.00; CAR=1.00; EF=1.00 | Normative screening |

The metric-separation layer tests whether conventional retrieval metrics reproduce procedural qualification across 219 packets with complete upstream precision, recall and F1 scores. They do not. Upstream recall correlated 0.06 with protocol-defined qualification; recall $\geq 0.8$ captured every qualified packet but produced 153 false positives, for precision 0.29. Bootstrap and permutation tests preserved the separation: the recall-threshold precision interval was 0.23–0.35, the high-recall blocked-rate interval was 0.65–0.77 and the recall/status permutation test returned $p = 0.354$. Across 13 simplified substitute rules and 3,252 predictions, recall, F1, precision, total score, source binding, authority-material coverage, review readiness and their combinations all failed to reproduce the full protocol. The best simplified rule still produced 38 false positives. Metrics may be useful; they are not status-conferring.

The issue-family generalization layer tests whether that result is an artefact of a particular legal issue family. It labels 242 issue-specific packets across U.S. agency deference after *Loper Bright*, English mesothelioma causation after *Fairchild*, EU GDPR Article 15 access rights, Canadian standard of review after *Vavilov*, and German/EU right-to-be-forgotten review. Each family is held out in turn. On the remaining families, the runner selects the best simplified substitute rule by F1 and the lowest-false-positive simplified rule; the selected rule is then evaluated on the held-out family. Across 3,014 holdout predictions, the full protocol produced 0 false positives and 0 false negatives. The best trained simplified rule produced 38 holdout false positives and erred in 5/5 held-out folds. The substitute failure is therefore cross-issue, not a peculiarity of one authority set.

The multi-axis holdout layer then tests whether that result depends on the selected split. It repeats trained-substitute holdout evaluation across six axes: issue family, validation suite, system role, jurisdiction profile, claimed status and output-origin class. Each fold selects simplified substitute rules only from the training strata before applying them to the held-out stratum. Across 19,235 multi-axis holdout predictions and 91,333 training predictions, the full protocol produced 0 false positives and 0 false negatives. The best trained simplified rules produced 510 holdout false positives and erred in 17/34 folds; the lowest-FP trained rules produced 530 false positives and erred in 19/34 folds. The failure of weaker rules is therefore not just cross-issue. It also survives changes in suite source, role, jurisdictional profile, claimed status and output origin.

The policy-family robustness layer tests whether the result depends on one arbitrary policy instantiation. It evaluates 12 variants that change thresholds, rank-salience windows, procedural source-tag classes, role caps and failure severity. Across 42,192 policy-family evaluations, no variant promoted a below-screening packet to high status; conservative variants demoted 42 high-status allocations; every simplified substitute family still erred in 12/12 variants; best simplified rules produced 449 cumulative false positives, while the full protocol remained 0 false positives and 0 false negatives.

The gate-ablation layer tests gate necessity across the 63 packets that reached normative screening or decision-support status. It removes the evidence packet, procedural source tags, high-authority materials, counter-materials, review gate or role entitlement; for the 12 decision-support packets, it also removes the decision-adoption reason. All 390 ablations fell below the corresponding procedural level. Qualification is not carried by an isolated issue packet or a high total score; it depends on the procedural chain as a whole.

The gate-contrast witness layer isolates Proposition 1 directly. Across the same 63 qualified packets, it created 390 witness pairs that preserved the score vector, upstream metrics and system role while changing one evidence, source-tag, high-authority, counter-material, jurisdiction, contestability or adoption predicate. All 390 pairs separated status: 378 fell to reference information and 12 decision-support witnesses fell to normative screening.

The source-chain attack layer applies the same claim at data level. For each of the 63 qualified packets, the harness generated every nonempty combination of five attacks: locator mismatch, output-source-link mismatch, nonprocedural source tags, high-authority omission and counter-material omission. All 1,953 attacked variants lost high status despite mean audit score 11.24 and mean upstream recall 0.99: 63 were downgraded, 378 were suspended and 1,512 were withdrawn; 441 were capped at reference information and 1,512 were withdrawn from external legal effect. A high-scoring output therefore cannot retain procedural status when the reconstructable legal-material chain is falsified.

The contestation-challenge layer makes the $T$ dimension dynamic. For the same 63 qualified packets, all 252 valid source, authority, jurisdiction and channel challenges blocked high status; all 63 unsupported controls preserved status. Contestability is an operational gate, not a ceremonial opportunity to object.

The metamorphic policy layer tests status-function invariants without reading scenario expected labels. It applies seven transformations across the 264 primary scenario packets: claim escalation never raises allowed status; perfect upstream metrics leave status invariant; a back-office role cap blocks external status; perfect scores plus an authorized role still cannot create a decision reason without adoption artefacts; non-procedural source tags and review-gate removal block high status; and benign procedurally tagged source additions preserve qualified status. All 1,233 metamorphic tests passed, showing that status allocation follows policy invariants rather than authored expected labels.

The policy-mutation layer killed 15/15 mutants across 3,111 evaluations. The mutants removed evidence-packet, procedural-source-tag, high-authority, counter-material, review, jurisdiction, role-cap, failure-cap and decision-adoption requirements, or treated recall, source binding, authority-material coverage, review posture, total score and model identity as status-conferring. The killed mutants produced 2,264 invalid promotions and 2 false negatives. This is the executable version of the doctrinal claim: the protocol is not a policy essay; it is a falsifiable status predicate.

The review-provenance layer ran 627 evaluations to test whether human-review language is merely decorative. Adding completed-review or authorized-adoption labels to incomplete packets did not supply the missing legal-material chain: 402/402 review/adoption placebos remained blocked. Removing provenance from qualified packets also behaved as required: 189/189 high-status provenance defects were blocked, and 36/36 decision-provenance defects were demoted. The test separates review labels from review records. Completed review must leave a contestability channel, jurisdiction assumption and adoption trail before it can carry procedural status.

The claim-anchor layer ran 1,080 mutations across 63 qualified packets, 250 output units and 290 source links. Removing proposition text or link-to-claim bindings downgraded 250/250 and 290/290 cases, respectively; removing support attestations or locators withdrew 290/290 and 250/250 cases. The layer blocks a common overclaim: a citation list is not a procedural evidence chain unless each material proposition is bound to a specific source anchor.

The repair-frontier layer applies the converse test to blocked high-status claims. Across 193 blocked claims and 5,097 counterfactual variants, all 193 were repairable by declared gate families. The minimal-frontier distribution was concentrated: 98 claims required one gate family, 50 required two, 41 required three, three required four and one required five. Source-binding, authority-packet, review-contestability and rank-salience completion dominated. Downgrade is therefore diagnostic: the protocol identifies the smallest missing artefact class before reconsideration.

The jurisdiction-profile layer tests profile-bound portability. The harness checked

233 high-status claims and found 251/251 profile checks supported. It then mutated all 63 qualified packets: replacing specific assumptions with the valid generic profile preserved status in 63/63 cases, removing the assumption downgraded 63/63 cases and substituting a mismatched profile downgraded 63/63 cases, for 189/189 profile mutations passed. The rule is portable because authority assumptions are explicit, parameterized and replayable.

The ranking-visibility layer makes salience an auditable status condition. Across 248 high-status claims, 956 rank-window visibility checks produced a counter-material visibility curve: 14/248 at rank 1, 84/241 by rank 2, top-3 counter-material visibility was 198/235, and 207/232 by rank 4. The median first counter-material rank was 3.0 and the mean reciprocal first-counter rank was 0.43. In the 85 packets with enough non-counter material to run a true rank-order intervention, the counterfactual moved counter-material below the visibility window while preserving authority coverage, counter-material recall, evidence fidelity and procedural source-tag coverage. All 85 were downgraded below normative screening, and drifted top-3 counter-material visibility was 0/85. The same authority set can be present while rank salience independently controls procedural status.

The certificate layer makes procedural status proof-carrying. Each packet records identity, hashes, full policy body, profile, scores, role cap, missing gates, failure cap, metrics, 17 proof obligations and a derivation hash. Replaying those certificates verified 264/264 certificates, 6,072/6,072 replay checks and 4,488/4,488 proof obligations. The tamper layer mutated certificate identities, policy bodies, scores, roles, gates, caps, metrics, obligations and certificate-set membership; all 5,811 tamper cases were rejected. A legal AI output can therefore carry its own procedural-status proof object.

The policy-constants replay layer is the independent executable specification. It reads `policy/legal_output_policy.json` for thresholds, status ranks, role caps, protected reliance gates, source-tag classes, jurisdiction-profile aliases and failure dispositions; then it recomputes score candidates, inferred roles, missing gates, metrics, failure flags, failure caps and final status without importing the harness model. It verified 264/264 packets and 4,752/4,752 replay checks. The status allocation must therefore be reproducible both as a proof-carrying certificate and as a second implementation parameterized by the same JSON constants.

The final layers test coding stability. Strict and lenient recoding preserved status for 262/264 packets, with weighted agreement of 1.00 against the strict pass and 0.98 against the lenient pass. Fixed-seed Monte Carlo perturbation of all six audit scores produced 66,000 perturbed evaluations, 0.937 sample-level status stability and 0.916 high-status stability among qualified packets. Score-blinded dual coding of 240 non-holdout packets reached 0.99 inter-coder exact agreement, 0.99 Cohen's kappa and 0.96 quadratic weighted kappa; the weakest dimension-level kappa was 0.93. Base-score calibration was tracked separately: the weakest base-dimension kappa was 0.37 for $Q$ under class imbalance, with 0.97 exact agreement. The coding result supports the same rule as the formal and adversarial layers: status is stable when the legal-material chain is stable, and instability concentrates at the intended threshold gates.

## 8.7 Construct-operationalization coverage

The validation matrix also tests whether the paper's theoretical constructs are actually operationalized. The coverage pass maps 10 core constructs–status ordering, role and failure caps, the six-part audit vector, evidence packets, authority hierarchy, counter-material salience, contestability and adoption, substitute-theory separation, architecture and identity portability, and replication/tamper resistance–to 31 committed evidence layers. Every construct is backed by at least 4 evidence layers and at least 3 evidence roles. The coverage script passed 40/40 construct-operationalization checks, so the construct map is executable rather than a prose crosswalk.

## 8.8 Threat-model coverage

The validation matrix also includes a threat-model coverage pass rather than treating validity as a prose limitation. It maps 8 validity threats–construct, internal, source-chain, external, architecture, policy, provenance and annotation validity–to 30 committed evidence layers. Every validity threat is backed by at least 4 evidence layers and at least 3 evidence roles. The coverage script passed 32/32 threat-model coverage checks, so each validity claim in the paper points to an executable layer rather than to a general disclaimer.

## 8.9 Worked application: Loper Bright evidence ladder

The model-output evidence ladder shows how the protocol treats one concrete case-recommendation task. The raw output `codex55-us-03` answered how legal research should treat *Skidmore* and *Kisor* after *Loper Bright*. It named *Loper Bright*, APA section 706, *Skidmore*, *Chevron* and *Kisor*; the committed scenario records upstream precision 0.95, recall 1.00 and F1 0.97. On ordinary retrieval terms, this looks strong. Under the status protocol, the raw answer remains reference information because the citations are not yet procedurally source-bound, the review gate is pending and no contestability channel is recorded.

The same visible answer qualifies differently when the legal-material chain changes. Once each output unit is linked to issue-manifest sources, locators, hashed support evidence and procedural source tags, and once review and contestability are completed, the output becomes normative material screening. If an authorized adopter then records human authorization, adoption reasons and a contestation record, the same source-bound chain may become a decision-support reason. If, however, the same chain is pushed into an irreversible external action without human authorization under an unaccountable external-disposition role, the output is withdrawn to no external legal effect. The answer's semantic content is not what moves it across statuses; the procedural record does. Table 16 summarizes the four status variants.

Table 16: Worked evidence ladder for *Loper Bright* output `codex55-us-03`

| **Variant** | Changed artefact | Allowed status | Interpretation |
|---|---|---|---|
| **Raw unverified** | Named authorities and high retrieval metrics; no procedural source bindings or contestability channel. | Reference information | Recall 1.00 and F1 0.97 do not create screening status. |
| **Contestable screening** | Manifest locators, hashed support evidence, procedural source tags, completed review and contestability channel. | Normative material screening output | The authority set becomes reconstructable and challengeable, but no decision adoption is recorded. |
| **Authorized decision support** | Human authorization, adoption reasons, contestation record and authorized decision-support role. | Decision-support reason | A screening output can enter reasons only through accountable adoption. |
| **Unauthorized external action** | Same source-bound chain, but irreversible external use without human authorization under an unaccountable role. | No external legal effect | Authorization and role caps can defeat an otherwise complete source chain. |

# 9 Implications

The framework has three implications for legal AI evaluation. First, model performance is not procedural permission. A system may meet technical retrieval benchmarks while failing procedural qualification because the output cannot be reconstructed, source-tagged or contested. Second, human oversight is not a status-conferring label. Oversight must leave review status, reasons, rejection records and adoption trails; a human signature without verifiable review should not upgrade an output's status. Third, accountability should follow control. Developers, deployers, professional users and institutional adopters control different risks and should carry corresponding explanation duties.

The framework sets the evaluation target for deployed systems: a legal AI output earns procedural status only by carrying the source chain, counter-material pathway, contestability route, adoption record and proof certificate required for that status. Public outputs, model outputs, source-bound repairs, adversarial variants, ablations, contestation challenges, replay checks, rank counterfactuals and coding tests all converge on that allocation rule.

# 10 Discussion and Limitations

The claim is a status-allocation claim, not a claim about general legal accuracy. The protocol decides when a visible legal-output packet may occupy a procedural position: reference information, professional support, normative material screening, decision-support reason or no external legal effect. It does not claim that a qualified packet is substantively correct on the merits of the underlying legal dispute. Merits review remains with the professional, party or authorized institution. This boundary is part of the contribution. It prevents strong retrieval performance, source labels or model fluency from being treated as permission to use an output procedurally.

The external-validity claim is likewise bounded but substantial. The validation matrix does not assert that five issue families exhaust legal retrieval. It tests whether the status predicate survives across common-law, civil-law, administrative-law and EU-rights profiles; public retrieval endpoints; raw and repaired model outputs; workflow architectures; provider identities; policy variants; issue-family holdouts; and multi-axis holdouts. Each new jurisdiction or deployment still has to instantiate its own authority hierarchy, counter-material set, source tags and adoption records. Once those artefacts are instantiated, the same predicate can be replayed. Generalization therefore attaches to the legal-material chain structure, not to any one corpus or model.

The strongest measurement risk is human coding of the base audit vector. The artifact addresses that risk without pretending it disappears. The harness computes status allocation, role caps, evidence-packet metrics, source-tag checks, claim-anchor checks, failure caps and certificate replay from committed packet fields; the initial 0–2 audit scores are coded evidence judgments. Strict and lenient recoding, fixed-seed score perturbation and score-blinded dual coding test whether the article's status results are fragile to those judgments. They show that disagreement concentrates near intended threshold boundaries, while high-status qualification still depends on source-chain, counter-material, contestability and adoption gates.

Finally, the artifact is a reproducible audit protocol rather than a production deployment study. The committed public snapshots and raw transcripts make the reported evaluation deterministic for review, while refreshed public endpoints may change because legal databases and model interfaces change. Such changes do not defeat the protocol; they create a new output evidence packet that must be frozen, scored and replayed. The article's contribution is therefore not a leaderboard result but a proof-carrying method for deciding which procedural status a case-recommendation output may claim.

# 11 Conclusion

Status follows auditability, not intelligence. That is the paper's central rule. A case recommendation system may be technically sophisticated, fluent or empirically strong and still have no procedural status if its legal-material chain cannot be reconstructed and challenged. Conversely, an output does not need to be a legal source or autonomous decision-maker to matter procedurally. Once it structures the visibility and ranking of authorities, it occupies the middle category of normative material screening.

The proposed framework makes that category operational. It separates output effect from system role and links both to source, corpus, ranking, counter-material, contestability, adoption and audit requirements. It allows an output to remain internal reference information, become professional support, qualify as normative material screening, or enter a decision reason only through accountable human adoption. It also supplies downgrade and withdrawal rules when the evidentiary and procedural chain fails.

The contribution is a stricter evaluation logic for legal AI outputs: no source chain, no status; no counter-material pathway, no screening qualification; no adoption record, no decision reason.

# Data and Code Availability

The companion artifact is part of the contribution. It contains the audit harness, scenario packets, public snapshots, raw model-output transcripts, validation scripts and generated Markdown/JSON reports. The primary replication command is `pythonscripts/run_full_validation.py`; it reruns the full validation matrix and writes the aggregate report under `experiments/full_validation/results/`. The committed public-source snapshots and raw model outputs make the reported results deterministic under local review; explicit refresh commands support fresh public collection.

# References

Ashley KD (1992) Case-based reasoning and its implications for legal expert systems. *Artificial Intelligence and Law* 1(2):113–208. https://doi.org/10.1007/BF00114920

Bench-Capon TJM, Sartor G (2003) A model of legal reasoning with cases incorporating theories and values. *Artificial Intelligence* 150(1–2):97–143. https://doi.org/10.1016/S0004-3702(03)00108-5

Binns R, Van Kleek M, Veale M, Lyngs U, Zhao J, Shadbolt N (2018) "It's reducing a human being to a percentage": Perceptions of justice in algorithmic decisions. In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, pp 1–14. https://doi.org/10.1145/3173574.3173951

Chalkidis I, Jana A, Hartung D, Bommarito M, Androutsopoulos I, Katz DM, Aletras N (2022) LexGLUE: A benchmark dataset for legal language understanding in English. In: *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics*. ACL, pp 4310–4330. https://doi.org/10.18653/v1/2022.acl-long.297

Citron DK, Pasquale F (2014) The scored society: Due process for automated predictions. *Washington Law Review* 89(1):1–33. https://digitalcommons.law.uw.edu/wlr/vol89/iss1/2/

Dahl M, Magesh V, Suzgun M, Ho DE (2024) Large legal fictions: Profiling legal hallucinations in large language models. *Journal of Legal Analysis* 16(1):64–93. https://doi.org/10.1093/jla/laae003

Dung PM (1995) On the acceptability of arguments and its fundamental role in nonmonotonic reasoning, logic programming and n-person games. *Artificial Intelligence* 77(2):321–357. https://doi.org/10.1016/0004-3702(94)00041-X

Gebru T, Morgenstern J, Vecchione B, Vaughan JW, Wallach H, Daumé H III, Crawford K (2021) Datasheets for datasets. *Communications of the ACM* 64(12):86–92. https://doi.org/10.1145/3458723

Goebel R, Kano Y, Kim MY, Rabelo J, Satoh K, Yoshioka M (2024) Overview and discussion of the Competition on Legal Information, Extraction/Entailment (COLIEE) 2023. *Review of Socionetwork Strategies* 18:27–47. https://doi.org/10.1007/s12626-023-00152-0

Gordon TF, Prakken H, Walton D (2007) The Carneades model of argument and burden of proof. *Artificial Intelligence* 171(10–15):875–896. https://doi.org/10.1016/j.artint.2007.04.010

Guha N, Nyarko J, Ho DE, Re C, Chilton A, Narayana A, Chohlas-Wood A, Peters A, Waldon B et al (2023) LegalBench: A collaboratively built benchmark for measuring legal reasoning in large language models. In: *Advances in Neural Information Processing Systems 36*. https://papers.nips.cc/paper_files/paper/2023/hash/89e44582fd28ddfea1ea4dcb0ebbf4b0-Abstract-Datasets_and_Benchmarks.html

Hou AB, Weller O, Qin G, Yang E, Lawrie D, Holzenberger N, Blair-Stanek A, Van Durme B (2025) CLERC: A dataset for U.S. legal case retrieval and retrieval-augmented analysis generation. In: *Findings of the Association for Computational Linguistics: NAACL 2025*. ACL, pp 7913–7928. https://doi.org/10.18653/v1/2025.findings-naacl.441

Kroll JA, Huey J, Barocas S, Felten EW, Reidenberg JR, Robinson DG, Yu H (2017) Accountable algorithms. *University of Pennsylvania Law Review* 165(3):633–705. https://scholarship.law.upenn.edu/penn_law_review/vol165/iss3/3/

Magesh V, Surani F, Dahl M, Suzgun M, Manning CD, Ho DE (2025) Hallucination-free? Assessing the reliability of leading AI legal research tools. *Journal of Empirical Legal Studies* 22(2):216–242. https://doi.org/10.1111/jels.12413

Mitchell M, Wu S, Zaldivar A, Barnes P, Vasserman L, Hutchinson B, Spitzer E, Raji ID et al (2019) Model cards for model reporting. In: *Proceedings of the Conference on Fairness, Accountability, and Transparency*. ACM, pp 220–229. https://doi.org/10.1145/3287560.3287596

Mökander J (2023) Auditing of AI: Legal, ethical and technical approaches. *Digital Society* 2:49. https://doi.org/10.1007/s44206-023-00074-y

Prakken H, Sartor G (2015) Law and logic: A review from an argumentation perspective. *Artificial Intelligence* 227:214–245. https://doi.org/10.1016/j.artint.2015.06.005

Raji ID, Smart A, White RN, Mitchell M, Gebru T, Hutchinson B, Smith-Loud J, Theron D, Barnes P (2020) Closing the AI accountability gap: Defining an end-to-end framework for internal algorithmic auditing. In: *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency*. ACM, pp 33–44. https://doi.org/10.1145/3351095.3372873

Rissland EL, Daniels JJ (1995) A hybrid CBR-IR approach to legal information retrieval. In: *Proceedings of the 5th International Conference on Artificial Intelligence and Law*. ACM, pp 52–61. https://doi.org/10.1145/222092.222125

Shao Y, Mao J, Liu Y, Ma W, Satoh K, Zhang M, Ma S (2020) BERT-PLI: Modeling paragraph-level interactions for legal case retrieval. In: *Proceedings of the Twenty-Ninth International Joint Conference on Artificial Intelligence*. IJCAI, pp 3501–3507. https://doi.org/10.24963/ijcai.2020/484

Turtle H (1995) Text retrieval in the legal world. *Artificial Intelligence and Law* 3:5–54. https://doi.org/10.1007/BF00877694

van Opijnen M, Santos C (2017) On the concept of relevance in legal information retrieval. *Artificial Intelligence and Law* 25:65–87. https://doi.org/10.1007/s10506-017-9195-8

Veale M, Van Kleek M, Binns R (2018) Fairness and accountability design needs for algorithmic support in high-stakes public sector decision-making. In: *Proceedings of the 2018 CHI Conference on Human Factors in Computing Systems*. ACM, pp 1–14. https://doi.org/10.1145/3173574.3174014

Wachter S, Mittelstadt B, Floridi L (2017) Why a right to explanation of automated decision-making does not exist in the General Data Protection Regulation. *International Data Privacy Law* 7(2):76–99. https://doi.org/10.1093/idpl/ipx005

Zheng L, Guha N, Anderson BR, Henderson P, Ho DE (2021) When does pretraining help? Assessing self-supervised learning for law and the CaseHOLD dataset of 53,000+ legal holdings. In: *Proceedings of the 18th International Conference on Artificial Intelligence and Law*. ACM, pp 159–168. https://doi.org/10.1145/3462757.3466088

Zheng L, Guha N, Arifov J, Zhang S, Skreta M, Manning CD, Henderson P, Ho DE (2025) A reasoning-focused legal retrieval benchmark. In: *Proceedings of the 4th ACM Symposium on Computer Science and Law*. ACM, pp 169–193. https://doi.org/10.1145/3709025.3712219